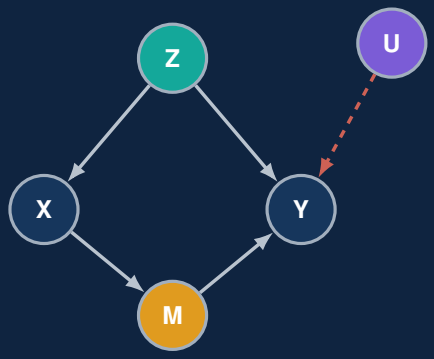


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 15

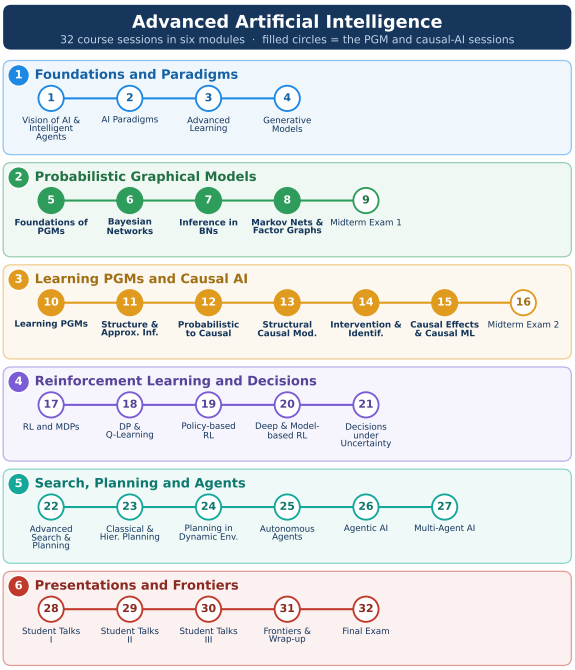
Causal Effects and Machine Learning

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 15 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, the core theory organized in parts, worked numerical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s15_examples.py` for the numbers, `python/s15_figures.py` for the figures), and `questions/s15_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions; this is session 15.

Contents

Session 15 Causal Effects and Machine Learning 1

15.1 The Potential-Outcomes Framework . . . 2
15.2 Estimands and Identification . . . 3
15.3 Outcome Regression, Matching and Stratification . . . 5
15.4 Propensity Scores and Inverse Propensity Weighting . . . 7
15.5 Doubly Robust Estimation . . . 8
15.6 Heterogeneous Effects and Meta-Learners . . . 9
15.7 Causal Forests and Honesty . . . 10
15.8 Double/Debiased Machine Learning . . . 11
15.9 Uplift Modelling and Policy Learning . . . 11
15.10 Off-Policy Evaluation in Reinforcement Learning . . . 12
15.11 Agentic AI: A/B Tests and Uplift Routing . . . 13
15.12 Multi-Agent AI: When SUTVA Fails . . . 14
15.13 Summary . . . 15

SESSION 15 ADVANCED ARTIFICIAL INTELLIGENCE

Causal Effects and Machine Learning

ATE, treatment effects, counterfactuals, causal effect estimation and causal ML

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I

Potential Outcomes & Estimands

SUTVA, ignorability, positivity; ATE, ATT, ATC, CATE, ITE; what is identified.

PART II

Estimating Average Effects

Regression, matching, stratification, propensity scores, IPW, AIPW.

PART III

Heterogeneity & Causal ML

Meta-learners, causal forests, double ML, uplift and policy learning.

PART IV

AI Connections

Off-policy evaluation, agent A/B tests, interference among agents.

LEARNING OBJECTIVES

- State the potential-outcomes framework (SUTVA, consistency, ignorability, positivity) and derive identification formulas for the ATE, ATT, ATC and CATE; explain why individual effects are only *bounded*.
- Build and compare the classical estimators: outcome regression, matching, stratification, Horvitz–Thompson and Hájek IPW, and the doubly robust AIPW estimator.
- Estimate heterogeneous effects with S-, T-, X-, R- and DR-learners and causal forests, and turn them into targeting policies evaluated by Qini curves.
- Explain **double/debiased machine learning**: Neyman orthogonality and cross-fitting.
- Recognise the same estimators in RL (importance-sampling and doubly robust off-policy evaluation), in A/B tests of agents, and in multi-agent systems where SUTVA fails.

Where we are. Sessions 5–11 (course sessions 5–8 and 10–11) built the probabilistic machinery; Session 12 (course 12) showed $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$; Session 13 (course 13) introduced SCMs and counterfactuals; Session 14 (course 14) decided *when* a causal quantity is identified. This final session turns identification into *estimation* with finite data and machine learning. Next, the course moves to reinforcement learning (course sessions 17–20), where today's estimators reappear as off-policy evaluation, and later to agentic and multi-agent AI (course sessions 26–27).

OPENING MOTIVATION

Session 14 told us exactly which quantity to compute — $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z)$ under the backdoor criterion — but every worked example assumed the relevant conditional distributions were simply known. Real data is finite, noisy, and high-dimensional: Z might be

dozens of covariates, not one categorical variable, and $P(y \mid x, z)$ is exactly the kind of function machine learning is built to estimate. This capstone session connects the identification theory of Sessions 12–14 to the practitioner's actual workflow: build estimators, using tools students already know (regression, classification, the reweighting ideas from Session 11's importance sampling), that target precisely the causal quantity Session 14 proved was well-defined.

PART I

Potential Outcomes and Estimands

What exactly do we want to estimate, and what can data reveal?

PART I · POTENTIAL OUTCOMES

15.1 The Potential-Outcomes Framework

Notation $T \in \{0,1\}$: treatment Z: covariates $Y(t)$: potential outcome $e(z) = P(T{=}1 \mid Z{=}z)$ $\mu_t(z) = \mathbb{E}[Y \mid T{=}t, Z{=}z]$ $\tau(z) = \mathrm{CATE}(z)$ n_1, n_0: numbers of treated and control units

Session 13 defined counterfactuals inside an SCM: $Y_x(u)$ is the value Y would take in the mutilated model M_x. Statistics and econometrics use the same object under the name *potential outcome* (Neyman–Rubin), with assumptions phrased directly on the potential outcomes instead of on a graph.

Definition 15.1 (Potential outcomes)**.** For a binary treatment, each unit i carries two potential outcomes $Y_i(1)$ (outcome if treated) and $Y_i(0)$ (outcome if untreated). The **individual treatment effect** is $\tau_i = Y_i(1) - Y_i(0)$. In SCM language $Y_i(t) = Y_{\mathrm{do}(T=t)}(u_i)$.

Definition 15.2 (Identifying assumptions)**.** **A1. SUTVA**: the potential outcomes of unit i depend only on i's own treatment (no interference) and there is a single version of each treatment level.

A2. Consistency: $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$ — the observed outcome is the potential outcome of the treatment actually received.

A3. Ignorability (unconfoundedness): $(Y(0), Y(1)) \perp\!\!\!\perp T \mid Z$.

A4. Positivity (overlap): $0 < e(z) < 1$ for every z with positive density.

Figure 15.1 shows why causal inference is a *missing-data problem*: the column τ_i is never observed. Ignorability says the missing half is "missing at random given Z"; positivity guarantees that both halves are observed somewhere near every z.

Remark 15.1 (Graphs and ignorability). If Z satisfies the back-door criterion in a causal DAG (Session 14), then $Y(t) \perp\!\!\!\perp T \mid Z$ (Pearl 2009, Ch. 11): the graph is how one *justifies*

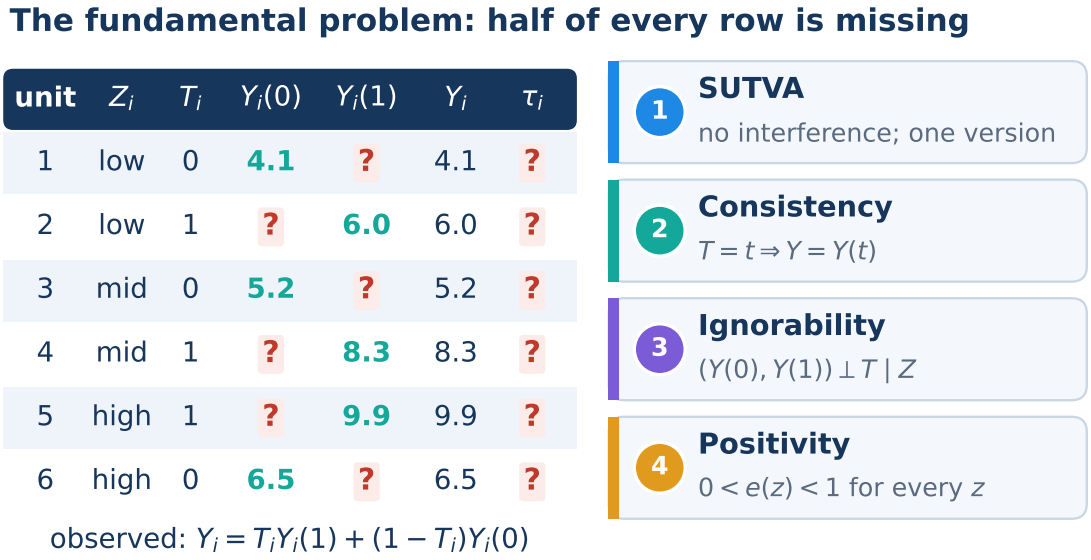


Figure 15.1. The fundamental problem of causal inference: every unit reveals one potential outcome, so every individual effect τ_i is missing. The four assumptions on the right are what turns the missing-data problem into an estimable one.

ignorability.

PART I · POTENTIAL OUTCOMES

15.2 Estimands and Identification

Suppose T is a binary treatment, Y the outcome, and Z a (possibly high-dimensional) set of covariates satisfying the backdoor criterion. The quantity of interest is usually the **average treatment effect**,

$$\mathrm{ATE} = \mathbb{E}\big[Y \mid \mathrm{do}(T{=}1)\big] - \mathbb{E}\big[Y \mid \mathrm{do}(T{=}0)\big] = \mathbb{E}_Z\Big[\mathbb{E}[Y \mid T{=}1, Z] - \mathbb{E}[Y \mid T{=}0, Z]\Big], \tag{15.1}$$

the second equality following directly from Session 14's backdoor adjustment theorem. But the ATE is only one of several estimands, each answering a different question (Table 15.1).

Definition 15.3 (Causal estimands)**.** With $\tau = Y(1) - Y(0)$: $\mathrm{ATE} = \mathbb{E}[\tau]$; $\mathrm{ATT} = \mathbb{E}[\tau \mid T{=}1]$ (effect on the treated); $\mathrm{ATC} = \mathbb{E}[\tau \mid T{=}0]$ (effect on the controls); $\mathrm{CATE}(z) = \mathbb{E}[\tau \mid Z{=}z]$; and the unit-level $\mathrm{ITE}_i = \tau_i$.

Theorem 15.1 (Identification under ignorability)**.** *Under SUTVA, consistency, ignorability and positivity,*

$$\mathbb{E}[Y(t)] = \mathbb{E}\big[\mu_t(Z)\big] = \mathbb{E}\Big[\frac{\mathbb{1}\{T{=}t\}\,Y}{P(T{=}t \mid Z)}\Big], \qquad t \in \{0,1\}, \tag{15.2}$$

$$\mathrm{CATE}(z) = \mu_1(z) - \mu_0(z), \qquad \mathrm{ATT} = \mathbb{E}[Y \mid T{=}1] - \mathbb{E}\big[\mu_0(Z) \mid T{=}1\big]. \tag{15.3}$$

The ATT needs only the weaker conditions $Y(0) \perp\!\!\!\perp T \mid Z$ and $e(z) < 1$.

Proof. For the regression form, $\mathbb{E}[Y(t) \mid Z] = \mathbb{E}[Y(t) \mid T{=}t, Z] = \mathbb{E}[Y \mid T{=}t, Z] = \mu_t(Z)$ by

ignorability and then consistency; conditioning on $\{T=t\}$ is legitimate because $P(T{=}t \mid Z) > 0$. Take expectations over Z. For the weighting form, $\mathbb{E}[\mathbb{1}\{T{=}t\}Y \mid Z] = \mathbb{E}[\mathbb{1}\{T{=}t\}Y(t) \mid Z] = P(T{=}t \mid Z)\,\mathbb{E}[Y(t) \mid Z]$, using consistency and then ignorability; divide by $P(T{=}t \mid Z)$ and average. The CATE follows by not averaging. For the ATT, $\mathbb{E}[Y(1) \mid T{=}1] = \mathbb{E}[Y \mid T{=}1]$ by consistency alone, and $\mathbb{E}[Y(0) \mid T{=}1] = \mathbb{E}\{\mathbb{E}[Y(0) \mid T{=}1, Z] \mid T{=}1\} = \mathbb{E}\{\mathbb{E}[Y(0) \mid T{=}0, Z] \mid T{=}1\} = \mathbb{E}[\mu_0(Z) \mid T{=}1]$, which uses only $Y(0) \perp\!\!\!\perp T \mid Z$ and $P(T{=}0 \mid Z) > 0$ on the support of $Z \mid T{=}1$. □

15.2.1 Unit-level counterfactuals: what even an RCT cannot reveal

The ITE τ_i depends on the *joint* distribution of $(Y(0), Y(1))$, while data — even a perfect randomized experiment — reveal only the two *marginals*. Hence the ITE, $\mathrm{Var}(\tau)$ and $P(\tau > 0)$ are not identified. For binary outcomes we can still bound them.

Proposition 15.1 (Fréchet bounds on benefit and harm)**.** *Let $p_1 = P(Y(1){=}1)$ and $p_0 = P(Y(0){=}1)$, so that $\mathrm{ATE} = p_1 - p_0$. The probability of benefit $P(Y(1){=}1, Y(0){=}0)$ is sharply bounded by $\max(0, p_1 - p_0) \le P(\textit{benefit}) \le \min(p_1, 1 - p_0)$, and the probability of harm is $P(Y(1){=}0, Y(0){=}1) = P(\textit{benefit}) - \mathrm{ATE}$. (These are the bounds on the probability of necessity and sufficiency of Session 13, read in potential-outcomes notation.)*

Proof. Write $P(a, b) = P(Y(1){=}a, Y(0){=}b)$ and let $b_{\text{ben}} = P(1, 0)$ be the benefit probability. The marginals force the other three cells of the 2×2 joint table: $P(1, 1) = p_1 - b_{\text{ben}}$ (from $P(1, 1) + P(1, 0) = p_1$), $P(0, 1) = p_0 - P(1, 1) = p_0 - p_1 + b_{\text{ben}}$ (harm), and $P(0, 0) = 1 - p_1 - P(0, 1) = 1 - p_0 - b_{\text{ben}}$. All four cells are ≥ 0 iff $b_{\text{ben}} \ge 0$, $b_{\text{ben}} \le p_1$, $b_{\text{ben}} \ge p_1 - p_0$ and $b_{\text{ben}} \le 1 - p_0$, which is the stated interval. Every value in that interval gives a valid joint distribution with the same marginals, so the bounds are sharp. Harm $= P(0, 1) = b_{\text{ben}} - (p_1 - p_0) = b_{\text{ben}} - \mathrm{ATE}$. □

Worked Example 15.1 (Two worlds with the same RCT)**.** A hypothetical trial with a binary outcome ($Y = 1$: recovery) gives $p_1 = 0.6$, $p_0 = 0.4$, so $\mathrm{ATE} = p_1 - p_0 = 0.6 - 0.4 = 0.20$. By Proposition 15.1, $P(\text{benefit}) \in [\max(0, 0.20), \min(0.6, 1 - 0.4)] = [0.20, 0.60]$ and $P(\text{harm}) = P(\text{benefit}) - 0.20 \in [0, 0.40]$. The two ends of the interval are two joint tables with the same marginals:

- *Monotone world* (benefit 0.20): harm $0.20 - 0.20 = 0$, always recover $P(1, 1) = 0.6 - 0.20 = 0.40$, never recover $P(0, 0) = 1 - 0.4 - 0.20 = 0.40$.
- *Polarised world* (benefit 0.60): harm $0.60 - 0.20 = 0.40$, always recover $0.6 - 0.60 = 0$, never recover $1 - 0.4 - 0.60 = 0$.

Check: in both worlds $P(Y(1){=}1) = P(1, 1) + P(1, 0) = 0.60$ and $P(Y(0){=}1) = P(1, 1) + P(0, 1) = 0.40$. Both worlds therefore produce identical trial data; deciding between them needs Session 13's structural assumptions (e.g. monotonicity, which forces the monotone world), not more data.

PYTHON LAB · `s15_examples.py` · `ex_probability_of_benefit()`

`P(benefit) in [0.20, 0.60], P(harm) in [0.00, 0.40]`; both tables give marginals $0.60, 0.40$.

Table 15.1. Estimands, their identification formulas and the assumptions each one needs.

Estimand	Identification formula	Needs
ATE	$\mathbb{E}[\mu_1(Z)-\mu_0(Z)] = \mathbb{E}\left[\frac{TY}{e(Z)} - \frac{(1-T)Y}{1-e(Z)}\right]$	full ignorability, $0 < e < 1$
ATT	$\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[\mu_0(Z) \mid T{=}1]$	$Y(0) \perp\!\!\!\perp T \mid Z$, $e < 1$
ATC	$\mathbb{E}[\mu_1(Z) \mid T{=}0] - \mathbb{E}[Y \mid T{=}0]$	$Y(1) \perp\!\!\!\perp T \mid Z$, $e > 0$
CATE(z)	$\mu_1(z) - \mu_0(z)$	ignorability and overlap at z
ITE	not identified; Fréchet bounds only	joint law of $(Y(0), Y(1))$

PART II

Estimating Average Treatment Effects

From identification formulas to estimators

PART II · AVERAGE EFFECTS

15.3 Outcome Regression, Matching and Stratification

Remark 15.2 (The running example: a confounded targeting study). Several results below are illustrated on one simulated study with known ground truth (`simulate_targeting()` in `s15_examples.py`, seed 2026). A firm has $n = 4000$ customers with engagement score $X \sim \text{Uniform}(0,1)$; X plays the role of the covariate Z. A targeted offer T is assigned with true propensity $e(x) = \text{sigmoid}(4x-2)$, where $\text{sigmoid}(u) = 1/(1+\exp(-u))$, so e ranges from $\text{sigmoid}(-2) = 0.119$ to $\text{sigmoid}(2) = 0.881$. The potential outcomes are

$$Y(0) = 1 + 2X + \varepsilon_0, \qquad Y(1) = Y(0) + (2 + 3X) + \varepsilon_1, \qquad \varepsilon_0, \varepsilon_1 \overset{\text{iid}}{\sim} N(0, 0.5^2),$$

hence $\mu_0(x) = 1 + 2x$, $\mu_1(x) = 3 + 5x$, $\tau(x) = \text{CATE}(x) = 2 + 3x$, $\sigma_0^2 = \text{Var}(\varepsilon_0) = 0.25$, $\sigma_1^2 = \text{Var}(\varepsilon_0 + \varepsilon_1) = 0.5$, and $\text{ATE} = \mathbb{E}[2 + 3X] = 2 + 3 \cdot 0.5 = 3.5$; in the drawn sample $\frac{1}{n}\sum_i (2 + 3X_i) = 3.506$ (the *sample* ATE). Engaged customers are both treated more often and have higher $Y(0)$, so X is a confounder. As a benchmark, the same customers also receive an independent coin-flip assignment $T^{\text{rct}} \sim \text{Bernoulli}(0.5)$ with outcome $Y(T^{\text{rct}})$. Nuisance models: $\hat{e}$ is a logistic regression of T on X; $\hat{\mu}_t$ come from one linear regression of Y on (X, T, XT) — both correctly specified. Monte Carlo studies redraw the data with $n = 1000$.

15.3.1 Outcome regression

Directly estimate $\hat{\mu}_1(z) = \widehat{\mathbb{E}}[Y \mid T{=}1, Z{=}z]$ and $\hat{\mu}_0(z) = \widehat{\mathbb{E}}[Y \mid T{=}0, Z{=}z]$ (any regression model — linear regression, gradient boosting, a neural network) using the observed data, and average their difference:

$$\widehat{\text{ATE}}_{\text{reg}} = \frac{1}{n}\sum_{i=1}^{n} \left[\hat{\mu}_1(z_i) - \hat{\mu}_0(z_i)\right]. \tag{15.4}$$

This is the most direct payoff of Session 10's learning machinery: it is ordinary supervised learning, applied twice (once per treatment arm), then combined via Session 14's adjustment formula. Its

weakness: if $\hat{\mu}_1, \hat{\mu}_0$ are misspecified (wrong functional form), the ATE estimate is biased, with no diagnostic built in. In the running example (Remark 15.2) it gives $\widehat{\text{ATE}}_{\text{reg}} = 3.522$.

COMMON PITFALL "Regress Y on T and Z and read off the coefficient of T" is *not* the ATE when effects are heterogeneous: it is a variance-weighted average of stratum effects.

15.3.2 Matching

Matching imputes each missing potential outcome by the observed outcome of a similar unit from the other arm. For the ATT, with $j(i) = \arg\min_{j:T_j=0} \|z_i - z_j\|$ the nearest control of treated unit i (controls may be reused: matching *with replacement*),

$$\widehat{\text{ATT}}_{\text{match}} = \frac{1}{n_1} \sum_{i:T_i=1} \big(Y_i - Y_{j(i)}\big).$$

Matching on the scalar $\hat{e}(z)$ (propensity-score matching) is justified by the balancing property below. With d continuous covariates the discrepancy $\|z_i - z_{j(i)}\|$ shrinks only like $n^{-1/d}$, so nearest-neighbour matching is biased in high dimension (Abadie and Imbens, 2006).

15.3.3 Stratification (subclassification)

Partition the propensity range into K strata (classically quintiles) and average within-stratum contrasts: $\hat{\tau}_{\text{strat}} = \sum_k (n_k/n)(\bar{Y}_{1k} - \bar{Y}_{0k})$, where n_k is the size of stratum k and $\bar{Y}_{tk}$ the mean outcome of its units with $T = t$. Five strata typically remove most confounding bias (about 90% for a single normally distributed covariate; Cochran, 1968); the rest comes from variation of $e(z)$ *within* strata.

Running example (Remark 15.2). Cutting $\hat{e}$ at its quintiles gives five strata of 800 customers each, so $n_k/n = 1/5$. The within-stratum differences are 2.390, 2.973, 3.533, 4.181, 4.789, whence $\hat{\tau}_{\text{strat}} = (2.390 + 2.973 + 3.533 + 4.181 + 4.789)/5 = 17.866/5 = 3.573$, against the sample ATE 3.506. The true mean CATEs of the five strata are 2.316, 2.917, 3.502, 4.093, 4.703: each stratum difference is slightly too large because $e(x)$ still increases with x inside each stratum — exactly the residual within-stratum confounding just described.

Worked Example 15.2 (Matching by hand)**.** Treated units (z, y): $(2, 9), (4, 12), (5, 13), (7.2, 16)$; controls: $(1, 5), (2, 6), (3, 7), (4.5, 9), (6, 10), (8, 12)$ (illustrative). Match each treated unit to the control with the smallest $|z_i - z_j|$, with replacement:

treated (z, y)	nearest control (z, y)	distance	runner-up distance	$y_i - y_{j(i)}$
$(2, 9)$	$(2, 6)$	0	1 (to $z = 1, 3$)	$9 - 6 = 3$
$(4, 12)$	$(4.5, 9)$	0.5	1 (to $z = 3$)	$12 - 9 = 3$
$(5, 13)$	$(4.5, 9)$	0.5	1 (to $z = 6$)	$13 - 9 = 4$
$(7.2, 16)$	$(8, 12)$	0.8	1.2 (to $z = 6$)	$16 - 12 = 4$

Hence $\widehat{\text{ATT}} = (3 + 3 + 4 + 4)/4 = 3.5$. The naive comparison uses the arm means $(9 + 12 + 13 + 16)/4 = 50/4 = 12.5$ and $(5 + 6 + 7 + 9 + 10 + 12)/6 = 49/6 = 8.167$, giving $12.5 - 8.167 = 4.333$; it is inflated because treated units have larger z, and z raises y.

PYTHON LAB · s15_examples.py · ex_matching_toy() and ex_ps_stratification()

```
matching ATT-hat = 3.500; naive = 12.500 - 8.167 = 4.333;  running  example:
stratum sizes [800 800 800 800 800]; within-quintile differences [2.39 2.973
3.533 4.181 4.789]; stratified ATE = 3.573.
```

PART II · AVERAGE EFFECTS

15.4 Propensity Scores and Inverse Propensity Weighting

Definition 15.4 (Propensity score)**.** $e(z) = P(T{=}1 \mid Z{=}z)$, the probability of receiving treatment given covariates.

Theorem 15.2 (Balancing property)**.** *$T \perp\!\!\!\perp Z \mid e(Z)$, and consequently $Y(0), Y(1) \perp\!\!\!\perp T \mid e(Z)$ whenever $Y(0), Y(1) \perp\!\!\!\perp T \mid Z$ (the backdoor / "no unmeasured confounding" condition, in potential-outcomes notation). Adjusting for the single scalar $e(Z)$ suffices — adjusting for all of Z is sufficient but not necessary.*

Proof. Since $e(Z)$ is a function of Z, $P(T{=}1 \mid Z, e(Z)) = e(Z)$, while by the tower property $P(T{=}1 \mid e(Z)) = \mathbb{E}[\mathbb{E}[T \mid Z] \mid e(Z)] = e(Z)$. The two agree, so $T \perp\!\!\!\perp Z \mid e(Z)$. For the second claim, write $\mathbf{Y} = (Y(0), Y(1))$: $P(T{=}1 \mid \mathbf{Y}, e(Z)) = \mathbb{E}[P(T{=}1 \mid \mathbf{Y}, Z) \mid \mathbf{Y}, e(Z)] = \mathbb{E}[e(Z) \mid \mathbf{Y}, e(Z)] = e(Z)$, which does not depend on $\mathbf{Y}$ (Rosenbaum and Rubin, 1983). □

The propensity score also suggests a second family of estimators, built on the idea of Session 11's importance sampling: we cannot sample from the "everyone treated" world directly, so we reweight the treated units we *did* observe to stand in for that counterfactual population.

Definition 15.5 (Inverse propensity weighting)**.** The Horvitz–Thompson (HT) and the normalised Hájek estimators are

$$\widehat{\mathrm{ATE}}_{\mathrm{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{T_i Y_i}{\hat{e}(z_i)} - \frac{(1-T_i)Y_i}{1-\hat{e}(z_i)}\right], \qquad \widehat{\mathrm{ATE}}_{\mathrm{Háj}} = \frac{\sum_i T_i Y_i/\hat{e}_i}{\sum_i T_i/\hat{e}_i} - \frac{\sum_i (1-T_i)Y_i/(1-\hat{e}_i)}{\sum_i (1-T_i)/(1-\hat{e}_i)},$$

with the shorthand $\hat{e}_i = \hat{e}(z_i)$. When the true e is known it can be used in place of $\hat{e}$.

Each treated unit is reweighted by $1/\hat{e}(z_i)$ — units who were *unlikely* to be treated given their covariates but were treated anyway are up-weighted, since they are rare and informative "surprises" standing in for many similar untreated units; this is precisely Session 11's importance-sampling weight $w = p/q$, with the proposal q here being "what treatment assignment actually looked like" and the target p being "what it would look like if everyone were treated". The weights of HT do not sum to n in a finite sample; Hájek divides by the realised sum, exactly like self-normalised importance sampling. After weighting, the covariate distribution of each arm matches that of the whole population.

Worked Example 15.3 (A hospital study: six estimates from one table)**.** A hypothetical observational study has three severity strata. Sicker patients are treated more often and recover less:

stratum	n_z	true $e(z)$	treated n_{1z}	controls n_{0z}	$\bar{Y}_{1z}$	$\bar{Y}_{0z}$	$\hat{\tau}_z = \bar{Y}_{1z} - \bar{Y}_{0z}$
mild	100	0.2	24	76	9	8	1
moderate	60	0.5	27	33	7	5	2
severe	40	0.8	33	7	5	2	3

In total $n = 200$, $n_1 = 24 + 27 + 33 = 84$ and $n_0 = 76 + 33 + 7 = 116$.
1 *Naive:* treated mean $(24 \cdot 9 + 27 \cdot 7 + 33 \cdot 5)/84 = (216 + 189 + 165)/84 = 570/84 = 6.7857$, control mean $(76 \cdot 8 + 33 \cdot 5 + 7 \cdot 2)/116 = (608 + 165 + 14)/116 = 787/116 = 6.7845$, difference $6.7857 - 6.7845 = 0.0012$ — confounding hides the effect almost completely.
2 *Stratification* (= saturated outcome regression): $(100 \cdot 1 + 60 \cdot 2 + 40 \cdot 3)/200 = 340/200 = 1.700$.
3 $ATT = (24 \cdot 1 + 27 \cdot 2 + 33 \cdot 3)/84 = 177/84 = 2.107$ and $ATC = (76 \cdot 1 + 33 \cdot 2 + 7 \cdot 3)/116 = 163/116 = 1.405$: treated patients are sicker and benefit more.
4 *HT with the true e:* $\hat{\mathbb{E}}[Y(1)] = \frac{1}{200}\left(\frac{24 \cdot 9}{0.2} + \frac{27 \cdot 7}{0.5} + \frac{33 \cdot 5}{0.8}\right) = \frac{1080 + 378 + 206.25}{200} = \frac{1664.25}{200} = 8.321$ and $\hat{\mathbb{E}}[Y(0)] = \frac{1}{200}\left(\frac{76 \cdot 8}{0.8} + \frac{33 \cdot 5}{0.5} + \frac{7 \cdot 2}{0.2}\right) = \frac{760 + 330 + 70}{200} = \frac{1160}{200} = 5.800$, so $\widehat{\text{ATE}} = 8.321 - 5.800 = 2.521$. The treated weights sum to $24/0.2 + 27/0.5 + 33/0.8 = 120 + 54 + 41.25 = 215.25 \neq 200$ (control weights: $76/0.8 + 33/0.5 + 7/0.2 = 95 + 66 + 35 = 196$) because by chance more mild patients were treated (24) than e predicts $(100 \cdot 0.2 = 20)$.
5 *Hájek:* $1664.25/215.25 - 1160/196 = 7.732 - 5.918 = 1.813$, much closer to 1.700.
6 With the *empirical* propensities $n_{1z}/n_z = 24/100, 27/60, 33/40 = 0.24, 0.45, 0.825$, HT and Hájek both equal 1.700 exactly (Worked Example 15.4).

PYTHON LAB · `s15_examples.py` · `ex_hospital_strata()`

```
naive 0.0012; strat 1.700; ATT 2.107; ATC 1.405; HT 2.521 (weights 215.25,
196.00); Hajek 1.813; empirical e:  1.700, 1.700
```

Worked Example 15.4 (Estimated propensities turn IPW into stratification — analytical)**.** If Z is discrete and $\hat{e}(z) = n_{1z}/n_z$, then for every stratum $\sum_{i:z_i=z} T_i Y_i = n_{1z}\bar{Y}_{1z}$, so $\sum_{i:z_i=z} T_i Y_i/\hat{e}(z) = n_{1z}\bar{Y}_{1z} \cdot n_z/n_{1z} = n_z\bar{Y}_{1z}$, and $\sum_{i:z_i=z} T_i/\hat{e}(z) = n_{1z} \cdot n_z/n_{1z} = n_z$. In the same way, with $1 - \hat{e}(z) = n_{0z}/n_z$, the control sums are $n_z\bar{Y}_{0z}$ and n_z. Hence both weight totals equal $\sum_z n_z = n$, and HT $= \frac{1}{n}\sum_z n_z(\bar{Y}_{1z} - \bar{Y}_{0z}) =$ Hájek $= \hat{\tau}_{\text{strat}}$. Estimating e — even when the true e is known — removes the chance imbalance seen in step 4. This is the finite-sample face of a deep fact: IPW with an *estimated* propensity is more efficient than with the true one (Hirano, Imbens and Ridder, 2003).

PART II · AVERAGE EFFECTS

15.5 Doubly Robust Estimation

Theorem 15.3 (Double robustness)**.** *Define the augmented IPW (AIPW) estimator*

$$\widehat{\text{ATE}}_{AIPW} = \frac{1}{n}\sum_i \underbrace{\left[\hat{\mu}_1(z_i) - \hat{\mu}_0(z_i) + \frac{T_i(Y_i - \hat{\mu}_1(z_i))}{\hat{e}(z_i)} - \frac{(1 - T_i)(Y_i - \hat{\mu}_0(z_i))}{1 - \hat{e}(z_i)}\right]}_{\hat{\varphi}_i}. \tag{15.5}$$

This estimator is consistent for the true ATE if either *the outcome models $\hat{\mu}_1, \hat{\mu}_0$ are correctly*

specified or *the propensity model $\hat{e}$ is correctly specified — it needs only one of the two to be right, not both.*

Proof sketch. Write $\hat{\mu}_1 = \mu_1 + \delta_1$ and let $\hat{e}$ be the fitted propensity. Using $\mathbb{E}[T \mid Z] = e(Z)$ and $\mathbb{E}[Y \mid T{=}1, Z] = \mu_1(Z)$ (ignorability), the treated-arm part has expectation $\mathbb{E}\big[\hat{\mu}_1(Z) + T(Y - \hat{\mu}_1(Z))/\hat{e}(Z)\big] = \mathbb{E}\big[\mu_1(Z) + \delta_1(Z)\big(1 - e(Z)/\hat{e}(Z)\big)\big]$. The error term vanishes if $\delta_1 \equiv 0$ (correct outcome model) or if $\hat{e} = e$ (correct propensity), leaving $\mathbb{E}[\mu_1(Z)] = \mathbb{E}[Y(1)]$; the control arm is symmetric (Robins, Rotnitzky & Zhao 1994). The bias of AIPW is thus a *product* of the two models' errors. The pseudo-outcomes $\hat{\varphi}_i$ are useful well beyond the ATE: averaging them over subgroups, regressing them on Z (DR-learner), or summing them over a policy's choices (policy learning) all inherit double robustness.

PART III

Heterogeneous Effects and Causal Machine Learning

From "does it work?" to "for whom, and whom should we treat?"

PART III · CAUSAL ML

15.6 Heterogeneous Effects and Meta-Learners

The ATE averages over the whole population, but a marketing team choosing whom to target, or a clinician choosing whom to treat, needs the **conditional average treatment effect**, $\mathrm{CATE}(z) = \mathbb{E}[Y \mid \mathrm{do}(T{=}1), Z{=}z] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0), Z{=}z]$. *Meta-learners* repurpose ordinary supervised learning models to estimate $\mathrm{CATE}(z)$:

- **T-learner**: fit two separate models $\hat{\mu}_1(z)$, $\hat{\mu}_0(z)$ (one per treatment arm, exactly as in outcome regression above) and set $\widehat{\mathrm{CATE}}(z) = \hat{\mu}_1(z) - \hat{\mu}_0(z)$.
- **S-learner**: fit a single model $\hat{\mu}(z, t)$ with treatment as an ordinary input feature, and set $\widehat{\mathrm{CATE}}(z) = \hat{\mu}(z, 1) - \hat{\mu}(z, 0)$.
- **X-learner**: impute individual-level "pseudo-effects" using the T-learner's cross-arm predictions, then regress these pseudo-effects on Z — generally more robust when treatment groups are very unbalanced in size. Concretely, $D_i^1 = Y_i - \hat{\mu}_0(z_i)$ for treated and $D_i^0 = \hat{\mu}_1(z_i) - Y_i$ for control units; regress D^1 on Z among the treated to get $\hat{\tau}_1$ and D^0 on Z among the controls to get $\hat{\tau}_0$, and combine $\hat{\tau}(z) = g(z)\hat{\tau}_0(z) + (1 - g(z))\hat{\tau}_1(z)$ with a weight $g \in [0, 1]$, usually $g = \hat{e}$: when few units are treated, $\hat{\tau}_1$ — built from $\hat{\mu}_0$, which is fitted on the large control arm — gets the larger weight $1 - \hat{e}$ (Künzel et al., 2019).
- **R-learner**: if $Y = \mu_0(Z) + T\tau(Z) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid T, Z] = 0$, then $m(z) = \mathbb{E}[Y \mid Z{=}z] = \mu_0(z) + e(z)\tau(z)$, and subtracting gives Robinson's decomposition $Y - m(Z) = (T - e(Z))\tau(Z) + \varepsilon$. Hence $\hat{\tau} = \arg\min_\tau \sum_i \big[(Y_i - \hat{m}(z_i)) - (T_i - \hat{e}(z_i))\tau(z_i)\big]^2 + \Lambda(\tau)$, where Λ is a regularisation penalty on the function τ (Nie and Wager, 2021).
- **DR-learner**: regress the AIPW pseudo-outcomes $\hat{\varphi}_i$ of (15.5) on z_i. With the true nuisances, the pseudo-outcome $\varphi^\circ = \mu_1(Z) - \mu_0(Z) + T(Y - \mu_1(Z))/e(Z) - (1 - T)(Y - \mu_0(Z))/(1 - e(Z))$ satisfies $\mathbb{E}[\varphi^\circ \mid Z] = \mu_1(Z) - \mu_0(Z) + 0 - 0 = \mathrm{CATE}(Z)$, because $\mathbb{E}[T(Y - \mu_1(Z)) \mid Z] =$

$e(Z)\big(\mu_1(Z) - \mu_1(Z)\big) = 0$ (and likewise for the control term); so any regression method applies.

Table 15.2. Meta-learners for $\mathrm{CATE}(z)$.

Learner	Strength	Weakness
S	simple; pools data; good when effects are small	regularisation shrinks $\widehat{\mathrm{CATE}}$ towards 0 or a constant
T	flexible; no shared shrinkage	small arm $\Rightarrow$ noisy; errors of two fits add
X	exploits the large arm; good for unbalanced designs	three stages; needs $\hat{e}$ for weighting
R	orthogonal loss; quasi-oracle rates	needs good $\hat{m}, \hat{e}$ and cross-fitting
DR	doubly robust pseudo-outcome	high variance where $\hat{e} \approx 0, 1$

Worked Example 15.5 (X-learner versus T-learner with 10% treated)**.** Simulate $n = 2000$ units (`xlearner_data()`, seed 11) with $Z \sim U(0,1)^5$, a complicated baseline $\mu_0(z) = 3\sin(\pi z_1 z_2) + 2(z_3 - 0.5)^2 + z_4$, a simple effect $\tau(z) = 1 + z_1$, treatment $T \sim \mathrm{Bernoulli}(0.1)$ independent of Z (the draw gives 210 treated, 1790 controls) and $Y = \mu_0(Z) + T\tau(Z) + \varepsilon$, $\varepsilon \sim N(0, 0.5^2)$. All base learners are random forests (150 trees, at least 5 units per leaf): the S-learner is one forest on (Z, T); the T-learner two forests, one per arm; the X-learner adds second-stage forests for D^1, D^0 and combines them with $g = e = 0.1$, i.e. $\hat{\tau} = 0.1\,\hat{\tau}_0 + 0.9\,\hat{\tau}_1$. Accuracy is measured on 1000 fresh test points (seed 12) by the root of the *precision in estimating heterogeneous effects*, $\sqrt{\mathrm{PEHE}} = \big(\frac{1}{1000}\sum_j (\hat{\tau}(z_j) - \tau(z_j))^2\big)^{1/2}$: it is 0.413 (S), 0.340 (T) and 0.152 (X). The T-learner must learn the complicated μ_0-shape twice, once from only 210 treated points; the X-learner borrows $\hat{\mu}_0$ from the 1790 controls and then only has to learn the simple τ.

PYTHON LAB · `s15_examples.py` · `ex_xlearner()`

```
S-learner:  root-PEHE = 0.413  T-learner:  0.340  X-learner:  0.152
```

PART III · CAUSAL ML

15.7 Causal Forests and Honesty

A **causal tree** partitions Z-space into leaves with $\hat{\tau}(\text{leaf}) = \bar{Y}_{1,\text{leaf}} - \bar{Y}_{0,\text{leaf}}$, choosing splits that maximise the heterogeneity of $\hat{\tau}$ across leaves rather than the accuracy of predicting Y. **Honesty**: one subsample chooses the splits, a disjoint one fills the leaves, so a split cannot sit where noise created a large apparent effect. A **causal forest** averages honest trees built on subsamples; its estimates are asymptotically normal and centred at $\mathrm{CATE}(z)$ (Wager and Athey, 2018). Honesty is the tree-level cousin of cross-fitting: *never evaluate a fit on the data that chose it.*

PART III · CAUSAL ML

15.8 Double/Debiased Machine Learning

Regularisation bias of flexible ML predictors leaks into naive plug-in estimates; double/debiased ML (Chernozhukov et al., 2018) fixes this with orthogonal scores and cross-fitting. Consider the **partially linear model**

$$Y = \theta T + g(Z) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid T, Z] = 0; \qquad T = m(Z) + v, \quad \mathbb{E}[v \mid Z] = 0. \tag{15.6}$$

Let $\ell(z) = \mathbb{E}[Y \mid Z{=}z]$. Taking $\mathbb{E}[\cdot \mid Z]$ of the outcome equation gives $\ell(Z) = \theta m(Z) + g(Z)$ (because $\mathbb{E}[T \mid Z] = m(Z)$ and $\mathbb{E}[\varepsilon \mid Z] = \mathbb{E}[\mathbb{E}(\varepsilon \mid T, Z) \mid Z] = 0$); subtracting, $Y - \ell(Z) = \theta\,(T - m(Z)) + \varepsilon$, so θ is the regression coefficient of the Y-residual on the T-residual (Frisch–Waugh–Lovell). The associated score $\psi = (Y - \ell(Z) - \theta(T - m(Z)))(T - m(Z))$ is **Neyman orthogonal**: small errors in $(\hat{\ell}, \hat{m})$ affect $\hat{\theta}$ only at second order.

Algorithm 15.1 DML with K-fold cross-fitting (partially linear model)

1: Split $\{1, \dots, n\}$ into folds $I_1, \dots, I_K$.
2: **for** $k = 1, \dots, K$ **do**
3: Fit $\hat{\ell}^{(-k)}$ and $\hat{m}^{(-k)}$ by any ML method on the data outside I_k.
4: For $i \in I_k$: $\tilde{Y}_i = Y_i - \hat{\ell}^{(-k)}(Z_i)$, $\tilde{T}_i = T_i - \hat{m}^{(-k)}(Z_i)$.
5: **end for**
6: $\hat{\theta} = \sum_i \tilde{T}_i \tilde{Y}_i / \sum_i \tilde{T}_i^2$; $\hat{\psi}_i = (\tilde{Y}_i - \hat{\theta}\tilde{T}_i)\tilde{T}_i$; $\widehat{\mathrm{se}} = \sqrt{\frac{1}{n}\sum_i \hat{\psi}_i^2} / \left(\frac{1}{n}\sum_i \tilde{T}_i^2\right) / \sqrt{n}$.

Worked Example 15.6 (DML in a partially linear model)**.** Simulate (15.6) (`plm_data()`) with $\theta = 0.5$, $Z \sim N(0, I_5)$, nonlinear $m(z) = \cos z_1 + 0.5 z_2^2 + 0.5 \tanh z_3$ and $g(z) = 2 \sin z_1 + z_2^2 + 0.5 z_4$, independent noises $v, \varepsilon \sim N(0,1)$, $n = 1000$, ten replications (seeds 300–309). Nuisances $\hat{\ell}, \hat{m}$ are random forests of 40 trees with fully grown leaves (one unit per leaf); DML uses $K = 2$ folds. Means of the estimates over the ten replications: OLS of Y on T alone 1.035; OLS on T and linear Z 1.053 (the quadratic confounding through z_2^2 is missed); partialling-out with *in-sample* forest residuals 0.458 (overfitted residuals shrink the signal); **cross-fitted DML** 0.488 (sd 0.040 across replications). In the first replication DML gives $\hat{\theta} = 0.457$ and, from Algorithm 15.1, s.e. 0.040, so the 95% CI is $0.457 \pm 1.96 \times 0.040 = 0.457 \pm 0.078 = [0.379, 0.535]$, which covers 0.5.

PYTHON LAB · `s15_examples.py` · `ex_dml_plm()`

```
OLS 1.035; OLS+Z 1.053; in-sample 0.458; DML 0.488 (sd 0.040); rep. 1:
0.457, CI [0.379, 0.535]
```

PART III · CAUSAL ML

15.9 Uplift Modelling and Policy Learning

A treatment rule $\pi : z \mapsto \{0, 1\}$ with per-unit cost c has net value $V(\pi) = \mathbb{E}[(\tau(Z) - c)\,\pi(Z)]$, maximised by the oracle rule $\pi^*(z) = \mathbb{1}\{\mathrm{CATE}(z) > c\}$. The plug-in rule "treat if $\widehat{\mathrm{CATE}}(z) > c$" is the bridge from causal ML to decisions; its value can be estimated from data by $\hat{V}(\pi) = \frac{1}{n}\sum_i \pi(z_i)(\hat{\varphi}_i - c)$ using the AIPW pseudo-outcomes, which inherits double robustness: its bias is a product of nuisance errors.

Qini / uplift curves evaluate a CATE *ranking* on held-out randomized data: sort units by decreasing $\widehat{\mathrm{CATE}}$; after targeting the top fraction ϕ, the Qini value is $Q(\phi) = \sum_{\text{top}} T_i Y_i - \sum_{\text{top}} (1 - T_i) Y_i \cdot n_T(\phi)/n_C(\phi)$, where the sums run over the targeted top fraction and $n_T(\phi), n_C(\phi)$ are the numbers of treated and control units in it: the control outcomes are rescaled to the size of the treated group, so $Q(\phi)$ is the incremental outcome generated. A useless model gives the straight line from 0 to $Q(1)$; the area between the curves is the Qini coefficient.

Worked Example 15.7 (Targeting with a cost)**.** In a hypothetical randomized campaign (`uplift_data()`, seed 21; $n = 6000$, the first 3000 for training, the last 3000 for testing), $Z \sim U(0,1)^2$, $T \sim \text{Bernoulli}(0.5)$, $Y = 1 + Z_2 + T\tau(Z) + N(0,1)$ with $\tau(z) = 2z_1 - 0.8$ (the offer *hurts* customers with $z_1 < 0.4$), and cost $c = 0.3$ in outcome units. A T-learner with one linear regression per arm gives $\widehat{\mathrm{CATE}}(z) = -0.709 + 1.846 z_1 + 0.007 z_2$. On the test half the Qini curve peaks at 552.7 when 62% are targeted, against 259.3 for targeting everyone. Policy values are computed on the 3000 test customers with their *true* $\tau(z)$ (possible only in a simulation) as $\frac{1}{3000}\sum_j (\tau(z_j) - 0.3)\pi(z_j)$. The policy "treat if $\widehat{\mathrm{CATE}} > 0.3$" treats 45.8% and earns +0.2014 per user — the same, to four decimals, as the oracle rule "treat if $\tau > 0.3$" on these customers (which treats 45.1%) — against −0.0951 for treating everyone. In the population the oracle treats exactly when $2z_1 - 0.8 > 0.3$, i.e. $z_1 > 0.55$, and earns

$$\mathbb{E}[(2Z_1 - 1.1)^+] = \int_{0.55}^{1} (2z - 1.1)\, dz = \left[z^2 - 1.1z\right]_{0.55}^{1} = (1 - 1.1) - (0.3025 - 0.605) = 0.2025.$$

PYTHON LAB · `s15_examples.py` · `ex_uplift_policy()`

```
Qini peak 552.7 at 62%; treat all -0.0951; CATE-hat > 0.3:  share 0.458,
+0.2014; oracle (test) 0.451, +0.2014; population oracle 0.2025
```

PART IV

Causal Effects in RL, Agents and Multi-Agent Systems

The same estimators under different names

PART IV · AI CONNECTIONS

15.10 Off-Policy Evaluation in Reinforcement Learning

A contextual bandit logs (x_i, a_i, r_i) under a behaviour policy $\pi_0(a \mid x)$; we want the value $V(\pi) = \mathbb{E}_x \mathbb{E}_{a\sim\pi}[r]$ of a new policy π *without deploying it*. This is causal effect estimation with action = treatment, context = covariates, and π_0 = a *known* propensity:

$$\hat{V}_{\text{IPS}} = \frac{1}{n}\sum_i \frac{\pi(a_i \mid x_i)}{\pi_0(a_i \mid x_i)} r_i, \qquad \hat{V}_{\text{DR}} = \frac{1}{n}\sum_i \Big[\sum_a \pi(a \mid x_i)\hat{r}(x_i, a) + \frac{\pi(a_i \mid x_i)}{\pi_0(a_i \mid x_i)}\big(r_i - \hat{r}(x_i, a_i)\big)\Big].$$

The self-normalised version is $\hat{V}_{\text{SNIPS}} = \sum_i w_i r_i / \sum_i w_i$ with $w_i = \pi(a_i \mid x_i)/\pi_0(a_i \mid x_i)$, and the direct method is $\hat{V}_{\text{DM}} = \frac{1}{n}\sum_i \sum_a \pi(a \mid x_i)\hat{r}(x_i, a)$ for a reward model $\hat{r}(x,a) \approx \mathbb{E}[r \mid x, a]$. IPS is HT, SNIPS is Hájek, DM is outcome regression and DR is AIPW (Dudík, Langford and Li, 2011). Positivity becomes *exploration*; in sequential RL the weights become products $\prod_t \pi(a_t \mid s_t)/\pi_0(a_t \mid s_t)$ whose variance grows with the horizon (Sutton and Barto, 2018, Ch. 5).

Worked Example 15.8 (Doubly robust off-policy evaluation)**.** Contexts x_1, x_2, x_3 are equally likely; actions are a_1, a_2, a_3; rewards are Bernoulli with success probabilities $r(x, a)$, and the logging policy $\pi_0(a \mid x)$ is known (`OPE` in the script):

	$r(x,a)$			$\pi_0(a \mid x)$			target
	a_1	a_2	a_3	a_1	a_2	a_3	$\pi(x)$
x_1	0.3	0.5	0.4	0.6	0.2	0.2	a_2
x_2	0.6	0.2	0.5	0.2	0.6	0.2	a_1
x_3	0.2	0.4	0.8	0.5	0.3	0.2	a_3

The logging policy is worth $\sum_a \pi_0(a \mid x) r(x, a) = 0.36, 0.34, 0.38$ in the three contexts (e.g. $0.6 \cdot 0.3 + 0.2 \cdot 0.5 + 0.2 \cdot 0.4 = 0.36$), on average 0.36; the deterministic target policy is worth $(0.5 + 0.6 + 0.8)/3 = 0.6333$. We simulate 1000 logs of $n = 300$ rounds each (seed 99). With a misspecified reward model that ignores the context ($\hat{r}(x, a)$ = mean logged reward of action a): DM has bias -0.2396 (RMSE 0.2415); IPS bias -0.0048, sd 0.0972; SNIPS bias -0.0061, sd 0.0598; DR bias -0.0076, sd 0.0655. (These small biases are shared Monte Carlo noise: all estimators use the same 1000 logs, and $0.0972/\sqrt{1000} = 0.003$.) With a saturated reward model ($\hat{r}(x, a)$ = the mean reward in cell (x, a)) DR coincides with DM in every log (sd 0.0592): for each context the residuals $r_i - \hat{r}(x_i, a_i)$ of the rounds that played the target action sum to zero, so the DR correction vanishes — the bandit analogue of Worked Example 15.4, where a saturated nuisance fit made two estimators coincide.

PYTHON LAB · `s15_examples.py` · `ex_ope_bandit()`

```
logging 0.3600; true 0.6333; DM bias -0.2396, RMSE 0.2415; IPS bias -0.0048,
sd 0.0972; SNIPS -0.0061, 0.0598; DR -0.0076, 0.0655; DR-good = DM-good sd
0.0592
```

PART IV · AI CONNECTIONS

15.11 Agentic AI: A/B Tests and Uplift Routing

Comparing two versions of an LLM agent on randomized traffic is an RCT, analysed by the difference in means with standard error $\sqrt{s_1^2/n_1 + s_0^2/n_0}$. With *logged* traces, where users or heuristics chose the version, task type and user segment confound the comparison and AIPW or DML with $Z =$ trace features is needed. The CATE answers: *for which queries is the expensive version worth it?*

Worked Example 15.9 (Routing between agent versions)**.** A hypothetical A/B test of agent v1 (single LLM call) against v2 (tool use + reflection, cost $c = 0.05$ success units per query) gives success rates v1/v2 of 0.80/0.82 (factual queries, 40% of traffic), 0.55/0.70 (reasoning, 35%) and 0.50/0.75 (coding, 25%). The CATEs (v2 minus v1) are $0.82 - 0.80 = 0.02$, $0.70 - 0.55 = 0.15$, $0.75 - 0.50 = 0.25$, and the ATE is $0.40 \cdot 0.02 + 0.35 \cdot 0.15 + 0.25 \cdot 0.25 = 0.008 + 0.0525 + 0.0625 = 0.1230$. Net value per query:

- always v1: $0.40 \cdot 0.80 + 0.35 \cdot 0.55 + 0.25 \cdot 0.50 = 0.32 + 0.1925 + 0.125 = 0.6375$;
- always v2: $0.40 \cdot 0.82 + 0.35 \cdot 0.70 + 0.25 \cdot 0.75 - 0.05 = 0.7605 - 0.05 = 0.7105$;
- route to v2 only where CATE $> c = 0.05$ (reasoning and coding): $0.40 \cdot 0.80 + 0.35 \cdot (0.70 - 0.05) + 0.25 \cdot (0.75 - 0.05) = 0.32 + 0.2275 + 0.175 = 0.7225$,

better than either fixed choice.

PYTHON LAB · s15_examples.py · ex_agent_routing()

```
CATE 0.02, 0.15, 0.25; ATE 0.1230; v1 0.6375, v2 0.7105, routing 0.7225
```

PART IV · AI CONNECTIONS

15.12 Multi-Agent AI: When SUTVA Fails

SUTVA's "no interference" clause is violated whenever units interact: a treated agent changes its neighbours' outcomes through communication, competition or shared resources. Write $Y_i = f(T_i, \bar{T}_{N(i)})$ with $\bar{T}_{N(i)}$ the share of treated neighbours (an *exposure mapping*). The *direct* effect $f(1,s) - f(0,s)$ differs from the *total* effect $f(1,1) - f(0,0)$ of treating everyone, which is what a deployment decision needs.

Worked Example 15.10 (Spill-overs among agents on a ring)**.** 400 agents sit on a ring, each interacting with two neighbours on each side, so $\bar{T}_{N(i)}$ is the share of treated among four neighbours; $Y_i = 1 + 1.0\,T_i + 1.5\,\bar{T}_{N(i)} + \varepsilon_i$ with $\varepsilon_i \sim N(0,1)$. The direct effect is 1.0 and the total effect is $f(1,1) - f(0,0) = 1.0 + 1.5 = 2.5$. Under independent Bernoulli(1/2) assignment, $\bar{T}_{N(i)}$ is independent of T_i with mean 1/2 in both arms, so the difference in means estimates only the direct effect, $1.0 + 1.5(\frac{1}{2} - \frac{1}{2}) = 1.0$: mean 0.991 over 500 replications (seed 5). Randomizing *blocks* of $m = 20$ adjacent agents (each block treated with probability 1/2) makes neighbours share treatment. In a treated block, the agent at each end has two of its four neighbours outside the block, each treated with probability 1/2, so $\mathbb{E}\bar{T}_{N(i)} = (2 + 2 \cdot \frac{1}{2})/4 = \frac{3}{4}$; the second agent from each end has one outside neighbour, $\mathbb{E}\bar{T}_{N(i)} = (3 + \frac{1}{2})/4 = \frac{7}{8}$; the other $m - 4$ agents have $\bar{T}_{N(i)} = 1$. The block average is $[(m-4) + 2 \cdot \frac{7}{8} + 2 \cdot \frac{3}{4}]/m = 1 - \frac{0.75}{m}$, and by symmetry it is $\frac{0.75}{m}$ in a control block. Hence $\mathbb{E}[\mathrm{DiM}] \approx 1 + 1.5\big(1 - \frac{1.5}{m}\big) = 1 + 1.5\big(1 - \frac{3}{2 \cdot 20}\big) = 1 + 1.5 \cdot 0.925 = 2.3875$, and the simulation gives mean 2.390. The remaining gap to 2.5 comes from the agents at block boundaries.

PYTHON LAB · s15_examples.py · ex_interference()

```
Bernoulli design:  mean 0.991;    cluster design (blocks of 20):  mean 2.390;
theory 2.3875
```

AI CONNECTION · Trustworthy AI

Counterfactual fairness (Kusner et al., 2017) asks that a prediction for individual i would be the same had i's protected attribute been different — a rung-3 counterfactual that, by Proposition 15.1, needs structural assumptions beyond the data. CATE estimates also expose *who* is harmed by a policy, a basic audit requirement; fairness audits and "the decision would change if…" explanations (course session 31) are causal effect estimates resting on today's four assumptions.

AI CONNECTION · Deep Learning and Generative Models

Two-headed networks such as TARNet (Shalit, Johansson and Sontag, 2017) learn a shared representation $\Phi(z)$ and outcome heads h_0, h_1 — a neural T-learner with an S-learner's data

sharing; adding a propensity head and an AIPW-style loss gives doubly robust training. Representation balancing penalises distance between $\Phi(Z) \mid T{=}1$ and $\Phi(Z) \mid T{=}0$, an echo of the balancing property (course sessions 3–4).

PART IV · AI CONNECTIONS

15.13 Summary

Table 15.3. Cheat-sheet of key formulas.

Object	Formula
Identification	$\mathbb{E}[Y(t)] = \mathbb{E}[\mu_t(Z)] = \mathbb{E}[\mathbb{1}\{T{=}t\}Y/P(T{=}t \mid Z)]$
ATT	$\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[\mu_0(Z) \mid T{=}1]$
AIPW score	$\varphi = \mu_1 - \mu_0 + \frac{T(Y-\mu_1)}{e} - \frac{(1-T)(Y-\mu_0)}{1-e}$
AIPW bias	$\mathbb{E}[(\hat{\mu}_1 - \mu_1)(\hat{e} - e)/\hat{e}]$ (product of errors)
Efficiency bound	$\mathbb{E}[\sigma_1^2/e + \sigma_0^2/(1-e) + (\tau - \mathrm{ATE})^2]$
DML (PLM)	$\hat{\theta} = \sum \tilde{T}\tilde{Y} / \sum \tilde{T}^2$, cross-fitted residuals
Policy value	$V(\pi) = \mathbb{E}[(\tau(Z) - c)\pi(Z)]$, optimum $\mathbb{1}\{\mathrm{CATE} > c\}$
Benefit bounds	$\max(0, p_1 - p_0) \le P(\text{benefit}) \le \min(p_1, 1 - p_0)$

THINK ABOUT IT An agent's logs record which tool it called and whether the user was satisfied. Which variables must you log so that ignorability can hold for "what if it always called tool A"?

THINK ABOUT IT Why can IPW with an estimated propensity be more precise than IPW with the true one?

THINK ABOUT IT If two agents learn from each other, what is the "unit" of a randomized experiment?

Further reading. Hernán and Robins (2020), Parts I–II; Imbens and Rubin (2015); Pearl (2009), Ch. 3, 11; Wasserman (2004), Ch. 8, 16; Rosenbaum and Rubin (1983); Cochran (1968); Abadie and Imbens (2006, 2008); Hirano, Imbens and Ridder (2003); Robins, Rotnitzky and Zhao (1994); Hahn (1998); Chernozhukov et al. (2018); Künzel et al. (2019); Nie and Wager (2021); Wager and Athey (2018); Dudík, Langford and Li (2011); Sutton and Barto (2018).

CONNECTIONS AND TAKEAWAYS

The notes' arc is now complete. Sessions 5–11 built a language for representing and computing with joint distributions — conditional independence, graphs, exact and approximate inference, and learning. Session 12 showed, with real numbers, that this language cannot by itself answer questions about intervention. Session 13 gave arrows a stronger, mechanistic meaning (SCMs) that supports exactly those questions. Session 14 showed when and how a causal

effect can be recovered from observational data and a graph alone (identification). And today closes the loop back to practice: once identification tells us *what* to estimate, the machine learning tools students already know — regression, classification, cross-fitting — tell us *how well* we can estimate it, and the special case of a randomized experiment shows that everything built since Session 12 was, all along, a way of solving — analytically, from observational data — the very problem that random assignment solves for free by design.

Course-level takeaway: causal inference is not a separate discipline bolted onto machine learning. It is machine learning aimed, by a graph-theoretic argument, at a precisely defined and provably estimable target — and knowing when that target is estimable at all, which this course's entire second half was devoted to establishing, is exactly the judgment a working data scientist needs before ever fitting a single model.

Next: course session 17 opens reinforcement learning, where action = treatment, behaviour policy = propensity and exploration = positivity.